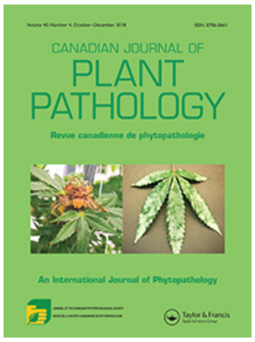


Taylor & Francis
Taylor & Francis Group

Canadian Journal of Plant Pathology

ISSN: 0706-0661 (Print) 1715-2992 (Online) Journal homepage: https://www.tandfonline.com/loi/tcjp20

# Flower and foliage-infecting pathogens of marijuana (*Cannabis sativa* L.) plants

Zamir K. Punja

**To cite this article:** Zamir K. Punja (2018) Flower and foliage-infecting pathogens of marijuana (*Cannabis sativa* L.) plants, Canadian Journal of Plant Pathology, 40:4, 514-527, DOI: 10.1080/07060661.2018.1535467

**To link to this article:** https://doi.org/10.1080/07060661.2018.1535467



Accepted author version posted online: 12 Oct 2018.
Published online: 16 Nov 2018.

*Can. J. Plant Pathol.*, 2018
Vol. 40, No. 4, 514–527, https://doi.org/10.1080/07060661.2018.1535467



Epidemiology/Épidémiologie

# Flower and foliage-infecting pathogens of marijuana (*Cannabis sativa* L.) plants

**ZAMIR K. PUNJA**

*Department of Biological Sciences, Simon Fraser University, 8888 University Drive, Burnaby, BC V5A 1S6, Canada*

(Accepted 18 September 2018)

**Abstract:** Flower buds of *Cannabis sativa* develop as inflorescences (buds) which are harvested and dried prior to sale. The extent to which fungal plant pathogens can colonize the buds prior to harvest has not been previously studied. Flower buds were sampled at various pre-harvest and harvest time periods during 2015–2017 at locations in British Columbia and Alberta to determine the range of fungi present. Isolated fungi were inoculated onto developing buds to determine the extent of tissue colonization. A pre- and post-harvest internal rot was associated with *Botrytis cinerea*, causing botrytis bud rot. In addition, two species of *Penicillium* – *P. olsonii* and *P. copticola* – were recovered from pre-harvest flower buds, as well as dried buds, and shown to cause penicillium bud rot. Scanning electron microscopy studies revealed colonization and sporulation on bracts and stigmas of the flower buds by *P. olsonii*. Several *Fusarium* species, which were identified using ITS rDNA sequences as *F. solani*, *F. oxysporum* and *F. equiseti*, were isolated from pre-harvest flower buds. These fungi colonized the flower buds following artificial inoculation and caused visible rot symptoms. The most severe symptoms were caused by *F. solani*, followed by *F. oxysporum* and, to a much lesser extent, *F. equiseti*. Powdery mildew infection of the foliage and flower buds was caused by *Golovinomyces*(*Erysiphe*) *cichoracearum*. The pathogen was detected on young vegetatively propagated cuttings and sporulation was abundant on older plants and on flower buds. The various fungi recovered from cannabis flower buds may be present as contaminants from aerially dispersed spores and have the potential to cause various types of pre- and post-harvest bud rot under conducive environmental conditions. Powdery mildew may be spread through aerially disseminated spores and infected propagation materials. Management of these pathogens will require monitoring of the growth environment for spore levels and implementation of sanitization methods to reduce inoculum sources.

**Keywords:** bud rot, cannabis, disease management, marijuana, pathogen detection, powdery mildew

**Résumé:** Les boutons floraux de *Cannabis sativa* forment des inflorescences qui sont récoltées et séchées avant d'être vendues. La mesure dans laquelle les agents pathogènes peuvent coloniser les inflorescences avant la récolte n'a pas été étudiée à ce jour. En 2015–2017, des boutons floraux ont été échantillonnés à différentes périodes précédant la récolte et durant la récolte, à quatre sites en Colombie-Britannique et en Alberta, pour évaluer la gamme de champignons qui s'y étaient développés. Les champignons isolés ont servi à inoculer des boutons émergents afin de déterminer l'ampleur de la colonisation des tissus. Une pourriture interne pré- et postrécolte a été associée à *Botrytis cinerea*, causant la pourriture grise des boutons. De plus, deux espèces de *Penicillium* — *P. olsonii* et *P. copticola* — ont été prélevées sur des boutons floraux avant la récolte, de même que sur des inflorescences séchées, et il s'est avéré qu'elles causaient aussi la pourriture des boutons. Des études faites au microscope électronique ont révélé que *P. Olsonii* avait colonisé les bractées et les stigmates des boutons floraux et qu'il y avait sporulé. Plusieurs espèces de *Fusarium*, qui ont été identifiées en tant que *F. solani, F. Oxysporum* et *F. equiseti*, en se basant sur les séquences de l'ITS de l'ADNr, ont été isolées à partir de boutons floraux d'avant la récolte. Ces champignons ont colonisé les boutons floraux à la suite d'une inoculation artificielle et ont causé des symptômes

Correspondence to: Zamir K. Punja. E-mail: punja@sfu.ca
All editorial decisions regarding this manuscript were made by James G. Menzies



Published online 16 Nov 2018

visibles de pourriture. Les symptômes les plus graves ont été causés par *F. solani*, suivi de *F. Oxysporum* et, dans une plus faible mesure, de *F. equiseti*. Une infection au blanc apparue sur le feuillage et les boutons floraux a été causée par *Golovinomyces*(*Erysiphe*) *cichoracearum*. L'agent pathogène a été détecté sur de jeunes boutures propagées végétativement et, sur les plants plus âgés et les boutons floraux, la sporulation était abondante. Les divers champignons collectés sur les boutons floraux de cannabis peuvent y être à titre de contaminants issus de spores dispersées par voie aérienne, et peuvent causer, quand les conditions environnementales sont propices, différents types de pourritures des boutons floraux avant la récolte et des inflorescences après la récolte. Le blanc peut être disséminé par des spores et du matériel de propagation infecté. La gestion de ces agents pathogènes requerra un suivi de l'environnement de croissance pour en évaluer la quantité de spores, ainsi que la mise en place de méthodes de désinfection pour réduire les sources d'inoculum.

**Mots clés:** blanc, cannabis, détection des agents pathogens, gestion des maladies, marijuana, pourriture des boutons

## Introduction

Cultivation of *Cannabis sativa* L., a member of the family Cannabeaceae which includes hemp and marijuana (referred to here as cannabis), has been practiced for centuries (Clarke, 1981; Small, 2017). Commercial production of industrial hemp for fibre and seed in Canada occurs in several provinces, including Alberta, Manitoba and Saskatchewan (Laate, 2012). The legalized growing and distribution of cannabis for medicinal and potentially recreational purposes is increasing in Canada and this increased cultivation is likely to cause an increase in previously undescribed pathogens that can negatively affect the production and quality of the crop. Recently, root rot pathogens affecting cannabis plants grown under hydroponic conditions (Punja & Rodriguez, 2018) as well as under field conditions (Punja et al., 2018) were described and they included species of *Pythium* and *Fusarium.*

While the pathogens affecting the foliage and flowers of hemp grown under field conditions have been described (McPartland, 1991, 1992), those potentially affecting cannabis plants grown indoors have not been extensively studied, due to restrictions placed on the cultivation of the psychoactive THC-containing plants in Canada and the USA. Different growing environments and cultivation methods are required for cannabis compared with hemp. Cannabis plants are propagated from cuttings that are rooted (requiring around 2–3 weeks) and then grown vegetatively for an additional 4–6 weeks, following which they are transferred to conditions of specific reduced lighting regimes to induce flowering (Small, 2017). Flower buds are harvested during the 12–14 week production cycle and dried to a moisture content of 8–10% (Small, 2017). They are then stored in vacuum-sealed bags or sealed plastic containers prior to distribution. Fungal colonization of flower buds generally occurs during the later stages of flower development and can be manifested as a pre-harvest or post-harvest bud rot (McPartland, 1992).

The objective of this study was to determine the prevalence of flower and foliar fungi affecting cannabis plants grown under greenhouse and controlled environmental conditions over a 3-year period (2015–2017). Following isolation and identification, pathogenicity studies were conducted to establish the extent to which the recovered microbes could induce disease symptoms. Preliminary results have been previously published (Punja & Rodriguez, 2017).

## Materials and methods

### *Sampling of plant materials*

Cannabis plants were sampled from within three greenhouse production facilities and one indoor production facility over the period from May 2015 to May 2017. The facilities were located in British Columbia (BC) and Alberta and were licensed under Health Canada regulations (Health Canada, 2013). The indoor facility was primarily hydroponic while cannabis plants were grown in cocofibre or soil in the three greenhouses. Flower buds at pre-harvest were collected randomly in replicates of five at different sampling times at 8–13 weeks of production, while buds at harvest were collected at 12 or 14 weeks, depending on the cannabis strain. Several different strains, defined as morphologically and genetically distinct plants represented by a unique name e.g. 'Pennywise', 'Girl Scout Cookies', 'Moby Dick', etc. (Punja et al., 2017) were sampled. In total, 60 samples (each with 5 buds) were collected during this study. A few buds showed initial symptoms of brown discolouration at harvest but most appeared visually healthy and therefore were not considered to be diseased (Fig. 1a). In addition, flower buds that were dried under commercially specified conditions prior to packaging were sampled (Fig. 1b). Leaves with visible infection by powdery mildew were also collected for pathogen identification from three locations in BC during 2016–2017.

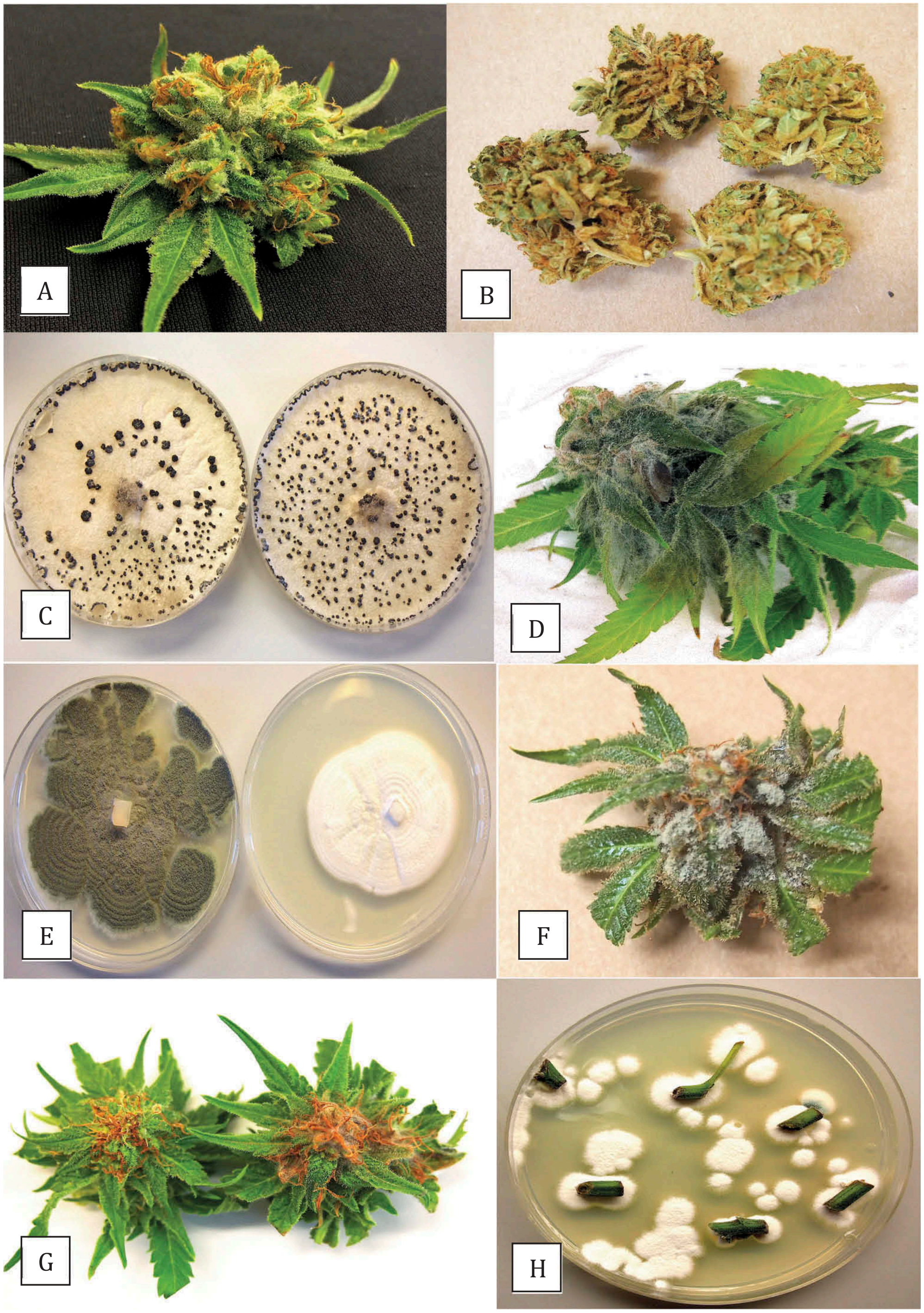


**Fig. 1** (Colour online) **(a)** Healthy fresh flower bud of *Cannabis sativa* (marijuana) harvested 10 weeks into the production cycle that was used for inoculation studies; **(b)** Dried mature flower buds at harvest; **(c)** Recovery of *B. cinerea* from naturally infected flower buds; **(d)** Symptoms of necrosis due to botrytis bud rot following artificial inoculation; **(e)** *Penicillium* species recovered from cannabis buds – (left) *P. olsonii*, (right) *P. copticola*; **(f)** Extensive colonization of flower bud by *P. olsonii*; **(g)** Slight necrosis following inoculation with *P. copticola* (right) compared with control (left); **(h)** Recovery of *P. copticola* from cannabis stem and petiole segments following surface-sterilization with sodium hypochlorite.

### *Isolation of fungi*

Flower buds were placed inside a plastic bag for 48 h at 23–25°C to allow any fungi to develop. Small tissue segments (~1 $mm^2$) were excised and surface-sterilized by dipping into 70% EtOH for 20 s, rinsed thrice in sterile water, blotted dry on sterile paper towels and then plated on potato dextrose agar (Sigma Chemicals, St. Louis, MO) with 100 mg $L^{-1}$ streptomycin sulphate (PDA). All dishes were incubated at ambient room temperature (23 ± 2°C) for 5–7 days. Emerging colonies were transferred to fresh PDA for subsequent identification to the genus level using general morphological criteria, including colony colour and size, and microscopic examination of spores. These isolations yielded cultures of *Botrytis*, *Penicillium* and *Fusarium* species. Hyphal tip sub-cultures were made and stored on PDA slants at 4°C and used for molecular identification.

### *Pathogen identification*

Cultures of representative isolates of *Botrytis*, *Penicillium* and *Fusarium* from buds, and powdery mildew samples, were sent to the University of Guelph Laboratory Services, Agriculture and Food Laboratory, Guelph, ON for species identification by PCR using the primers ITS1F-ITS4 (ITS1-F 5′-CTTGGTCATTTAGAGGAAGTAA-3′ and ITS4 5′-TCCTCCGCTTATTGATATGC-3′). The resulting sequences were compared to the corresponding ITS1-5.8S-ITS2 sequences from the National Center for Biotechnology Information (NCBI) GenBank database. Multiple sequence alignment of the respective isolates of each species was done using the CLUSTAL W program (http://www.genome.jp/tools/clustalw). The ITS sequences of *Fusarium solani* and *Golovinomyces* sp. were included in a phylogenetic analysis using the neighbour-joining (NJ) method (Saitou & Nei, 1987; Tamura et al., 2004) in the software MEGA v. 5 (Tamura et al., 2011). A bootstrap consensus tree was inferred from 1000 replicates to represent the distance. Branches corresponding to partitions reproduced in less than 50% bootstrap replicates were collapsed. The percentages of replicate trees in which the associated taxa clustered together in the bootstrap test (1000 replicates) were shown next to the branches as described by Felsenstein (1985). The outgroup used was *Sclerotinia sclerotiorum*.

GenBank accession numbers for the isolates recovered in this study are indicated in the Results.

### *Pathogenicity tests*

Flower buds were harvested from flowering plants in the 9–12 week growth cycle (Fig. 1a), visually examined to ensure they had no brown discolouration, and placed individually in Ziploc bags or baby food jars (200 mL) lined with moistened paper towels. For *B. cinerea*, an 8-mm diameter mycelial plug taken from the margin of 10-day-old PDA cultures was placed onto each flower bud (mycelial side down) and the bags were sealed. Controls received a sterile PDA plug. There were 10 replicates of each isolate to be tested. After 10 days of incubation at ambient room temperature (23–25°C), the buds were examined for symptoms of bud decay and evidence of fungal growth. A rating scale of 0–4 was used, where 0 = no visible mycelium on the bud; 1 = mycelium localized and covering < 25% of the surface; 2 = mycelium visible on bud surface, no rot; 3 = mycelium visible on bud surface, minor rot; and 4 = mycelium covering the bud surface, with extensive rot. Small segments of tissues were plated directly onto PDA to recover the pathogen. The experiment was conducted twice.

For *Penicillium* species, the mycelial plug inoculation method did not produce consistent results (data not shown). Therefore, a spore suspension was made by flooding a 2-week-old colony on PDA with 15 mL of sterilized distilled water, gently scraping the surface of the agar, and diluting this to 250 mL after adding 0.02% Tween 20. This yielded a spore suspension of ~$10^9$ spores per mL as determined using a hemocytometer. The buds were placed in individual baby food jars lined with moistened paper towels and atomized once with the spore suspension (about 0.2 mL). Controls received sterile distilled water with Tween 20. The extent of bud decay was assessed after 18 days of incubation at ambient room temperature and light conditions using the 0–4 scale described above. Small segments of tissues were plated directly onto PDA to recover the pathogens. The experiment was conducted three times, with 10 replicates per isolate tested.

For *Fusarium* species, cultures grown on PDA for 2 weeks were flooded with 15 mL of sterile distilled water, spores were dislodged using a glass rod, and the spore suspension diluted with distilled water to $10^4$ spores per mL as determined with a hemocytometer. Each flower bud received 200 uL of spore suspension as a droplet and placed individually after inoculation inside baby food jars lined with moistened paper towels. The extent of bud decay was assessed after 7 days of incubation at ambient room temperature and light conditions using a 0–4 scale described previously. The experiment was conducted twice, with 10 replicates per isolate tested. Inoculated tissues were plated to recover the pathogen.

Data from all experiments were averaged and the standard deviation was calculated from the replications and repetitions of each experiment.

*Scanning electron microscopy of fungal colonization of buds*

Small segments of bract tissue (0.5 × 0.5 cm) surrounding flower buds that had been inoculated with *P. olsonii* or *F. oxysporum*, as well as naturally infected powdery mildew infected bracts, were prepared for scanning electron microscopy as follows. Tissue segments were adhered to a stub using a graphite-water colloidal mixture (G303 Colloidal Graphite, Agar Scientific, UK) and Tissue-Tek (O.C.T. Compound, Sakura Finetek, NL). The sample was submerged in a nitrogen slush for 10–20 s to rapidly freeze it. After freezing, the sample was placed in the preparation chamber of a Quorum PP3010T cryosystem attached to a FEI Helios NanoLab 650 scanning electron microscope (Dept. of Chemistry, 4D Labs., Simon Fraser University). The frozen sample was sublimed for 5 min at −80°C, after which a thin layer of platinum (10 nm thickness) was sputter-coated onto the sample for 30 s at a current of 10 mA. The sample was moved into the SEM chamber and the electron beam was set to a current of 50 pA at 3 kV. Images were captured at a working distance of 4 mm, at a scanning resolution of 3072 × 2207 collected over 128 low-dose scanning passes with drift correction.

## Results

*Sampling of plant materials and pathogen identification*

Among almost 300 flower bud samples collected in this study, 6 (2%) showed visible brown rot that was attributed to *B. cinerea* and 11 (3.6%) samples had internal decay that was attributed to *F. solani* based on recovery of the respective fungi on PDA. The remaining samples were seemingly healthy but the following fungi and frequency of recovery from buds were recorded after surface-sterilization and plating tissues onto PDA: *F. oxysporum* (2.7% of samples), *F. equiseti* (0.9% of samples), *P. olsonii* (7.4% of samples) and *P. copticola* (10.6% of samples). Among the remaining samples, 0.3% had *Aspergillus niger* and the rest had no fungi. More than one fungal species was seldom found on the same sample. Samples with the highest number of fungi originated from greenhouses in which plants were grown in soil, but *F. oxysporum*, *P. olsonii* and *P. copticola* were also recovered from plants grown hydroponically at lower frequencies (data not shown).

*Pathogenicity tests*

Inoculation of flower buds with *B. cinerea* (Fig. 1c) resulted in rapid colonization by mycelium and the tissues were destroyed within 7–10 days, resulting in rotting and grey sporulation of the pathogen with a disease severity rating of 4 (Table 1, Fig. 1d). The pathogen was recovered from symptomatic tissues and formed characteristic masses of grey conidia and black irregularly shaped sclerotia developed on PDA (Fig. 1c). The isolate (GenBank accession no. MH782039) showed 100% sequence homology in the ITS1-ITS2 region to *B. cinerea* KF859918.1 (from forest species) and KX889115.1 (from strawberry fruits). Inoculation with the two *Penicillium* species (Fig. 1e) produced a slower developing dry rot (requiring up to 18 days) and pathogen sporulation was greenish-white in colour (Fig. 1f). *Penicillium copticola* (GenBank accession no. MH782038) was a much less aggressive species compared with *P. olsonii*, producing a disease severity value of 1 compared with 2.9 (Table 1, Fig. 1g), respectively, following artificial inoculation. External bud colonization by *P. olsonii* isolates (GenBank accession nos. MH782040, 782041, 782042) was extensive, reaching up to 80%. Control buds receiving sterile distilled water had background fungal contamination at a frequency of around 5%, represented mostly by *Penicillium* spp.

Scanning electron microscopic studies showed prolific production of conidia of *P. olsonii* and conidiophores developed on the surface of inoculated buds (Fig. 2a,b), as well as on stigmatic hairs where conidia could be observed forming in chains (Fig. 2c,d). Recovery of *P. copticola* and *P. olsonii* was confirmed from plated tissues. Interestingly, the former species was also recovered from stem and petiole segments of commercially grown plants that had not been inoculated (Fig. 1h) after surface-sterilization

**Table 1.** Extent of decay on cannabis flower buds following inoculation with six fungal species recovered from *Cannabis sativa.*

| Fungal species inoculated | Extent of bud decay (0–4 scale) |
|---|---|
| *Botrytis cinerea* | 4.0 ± 0.5 [a] |
| *Fusarium solani* | 3.5 ± 0.4 [b] |
| *Penicillium olsonii* | 2.9 ± 0.2 [c] |
| *Fusarium oxysporum* | 2.1 ± 0.3 [b] |
| *Fusarium equiseti* | 1.0 ± 0.1 [b] |
| *Penicillium copticola* | 1.0 ± 0.1 [c] |

Data represent the means from 10 buds. Each experiment was repeated at least twice. Inoculations were made using a mycelial plug (*B. cinerea*) or spore suspensions.
[a] Rated after 10 days of incubation at 23°C.
[b] Rated after 7 days of incubation at 23°C.
[c] Rated after 18 days of incubation at 23°C.
Rating scale used was: 0 = no visible mycelium on the bud; 1 = mycelium localized and covering < 25% of the surface; 2 = mycelium visible on bud surface, no rot; 3 = mycelium visible on bud surface, minor rot; and 4 = mycelium covering the bud surface, with extensive rot.

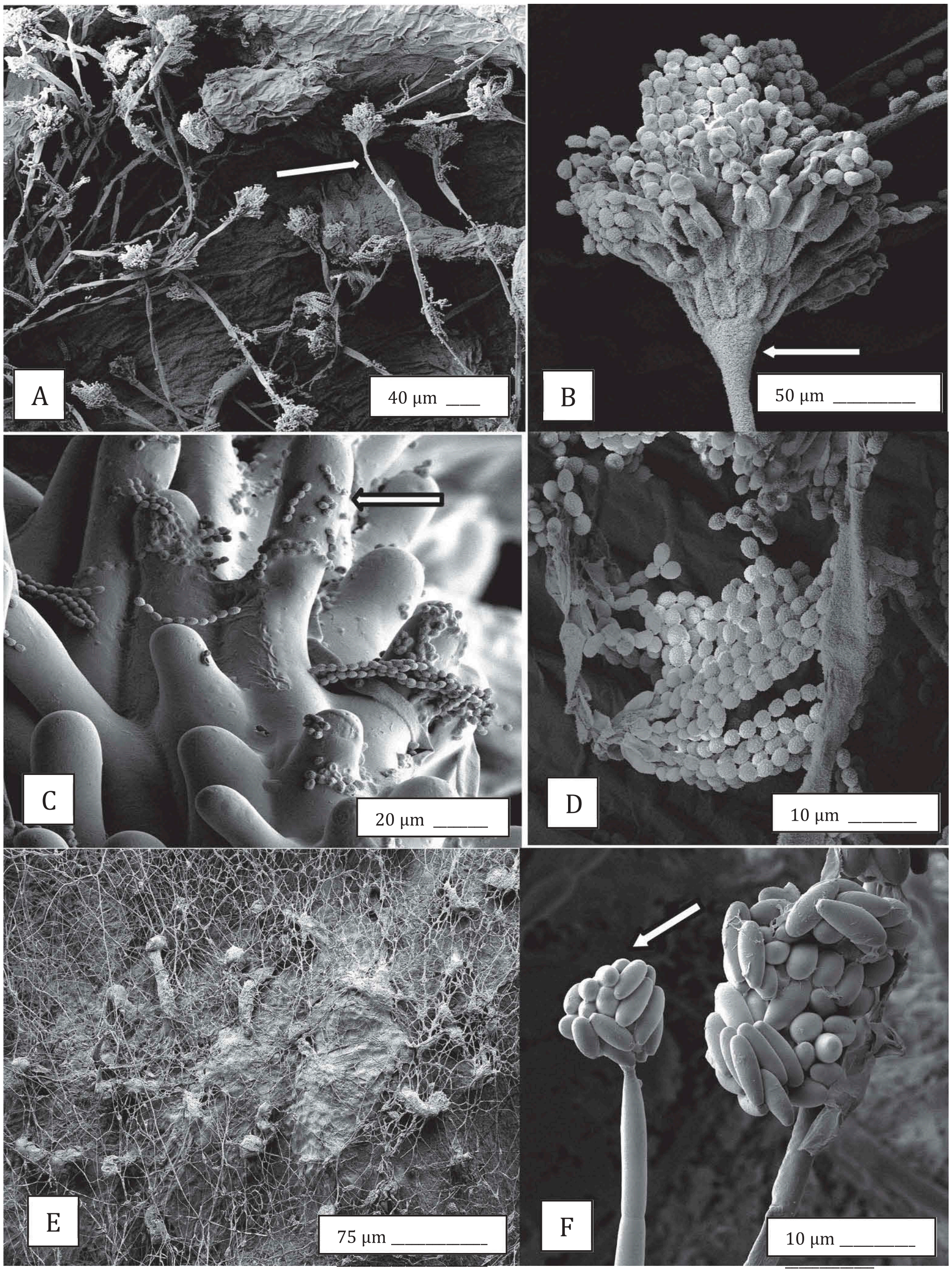


**Fig. 2** Scanning electron microscopy images of the development of *Penicillium olsonii* **(a–d)** and *Fusarium oxysporum* **(e, f)** on cannabis bud tissues following inoculation. **(a)** Conidiophores of *P. olsonii* (arrow) developing on the surface of the bract tissue; **(b)** Close-up of conidiophore (arrow) showing characteristic chains of spores of *Penicillium* produced from phialdes; **(c)** Spore chains stuck to the surface of stigmatic hairs (arrow); **(d)** Close-up of chains of spores; **(e)** Mycelium of *F. oxysporum* growing over surface of the bract tissues of buds; **(f)** Microconidia produced in clusters (arrow) on the bract tissue.

with 0.625% NaOCl for 3 min and plating onto PDA. The samples of dried and stored buds showed that *P. olsonii* was also present on these tissues and could be recovered following plating on PDA (data not shown).

Inoculation of flower buds with several *Fusarium* species revealed that the most extensive necrosis and mycelial growth was caused by *F. solani* (Fig. 3a,b), a moderate rate of infection was caused by *F. oxysporum* (Fig. 3c,d) and only slight necrosis was caused by *F. equiseti* (Fig. 3e,f).

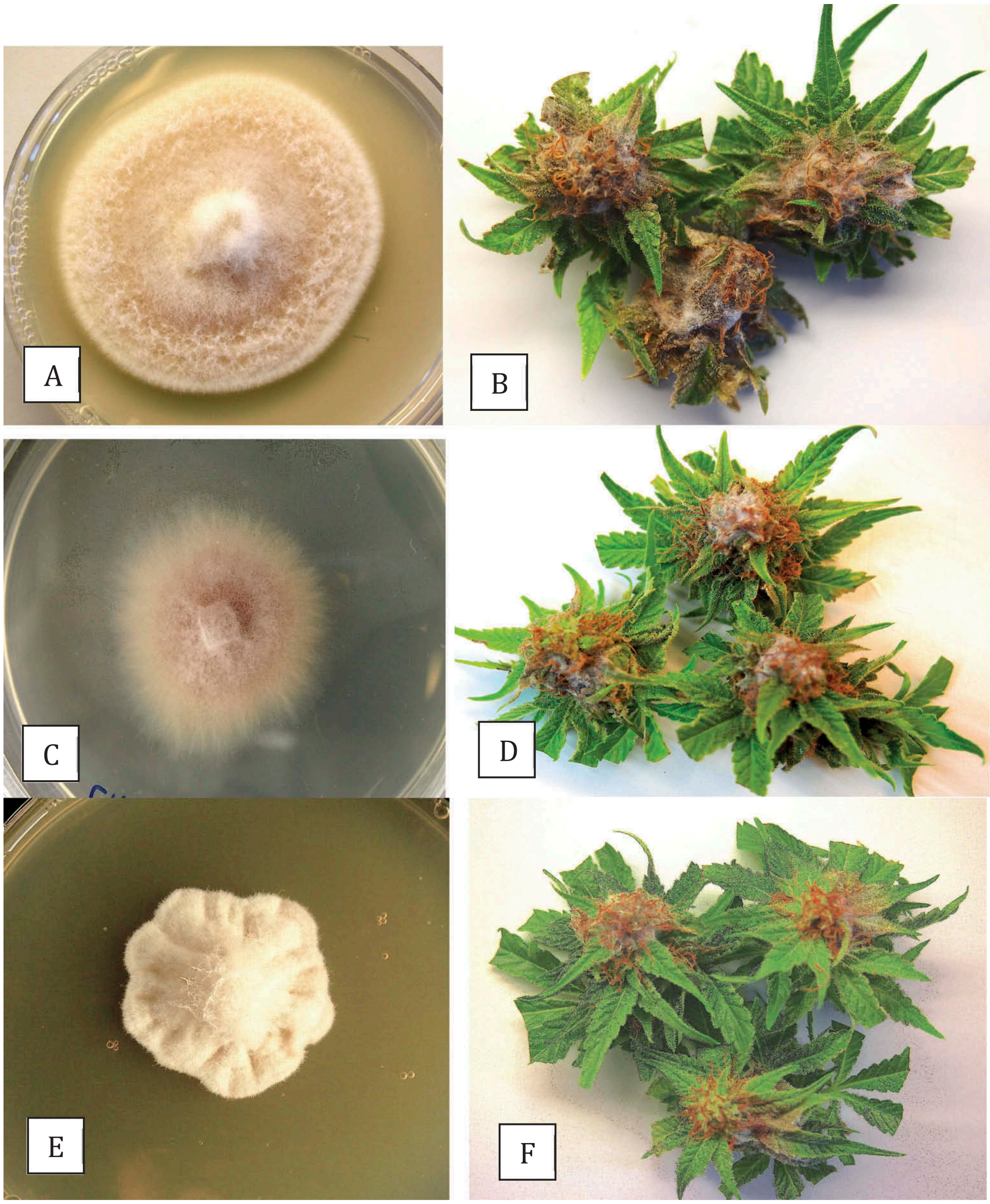


**Fig. 3** (Colour online) Colony morphology on PDA and results from pathogenicity tests of *Fusarium* spp. recovered from cannabis buds. **(a, b)** *F. solani*; **(c, d)** *F. oxysporum*; **(e, f)** *F. equiseti*. Photographs of inoculated buds were taken after 7 days of incubation at 23°C.

Scanning electron microscopy of *F. oxysporum*-infected buds showed extensive mycelial growth over the bud surface (Fig. 2e) as well as production of microconidia in clusters (Fig. 2f). Phylogenetic analysis of the *F. solani* isolates originating from flower buds (BC-3 and BC-4, GenBank accession nos. MH782034 and MH782035) showed that they formed a separate subgroup from the *F. solani* isolates (BC-1 and BC-2) previously recovered from diseased roots (Punja & Rodriguez, 2018) but all isolates from cannabis plants were grouped with a range of isolates from different hosts, including legumes from Iran and medicinal plants from China (Fig. 4).

*Powdery mildew*

White mycelium characteristic of powdery mildew developed on very young leaflets and older leaves (Fig. 5a–d), on young shoots, and on buds of cannabis plants. Scanning electron microscopy studies of mildew-infected cannabis leaves showed the conidia were formed singly and in chains on conidiophores developing directly from the mycelium (Fig. 6a,b). The foot cells of the conidiophore were distinctly curved (Fig. 6c). On infected flower buds, powdery mildew mycelium development and spore production

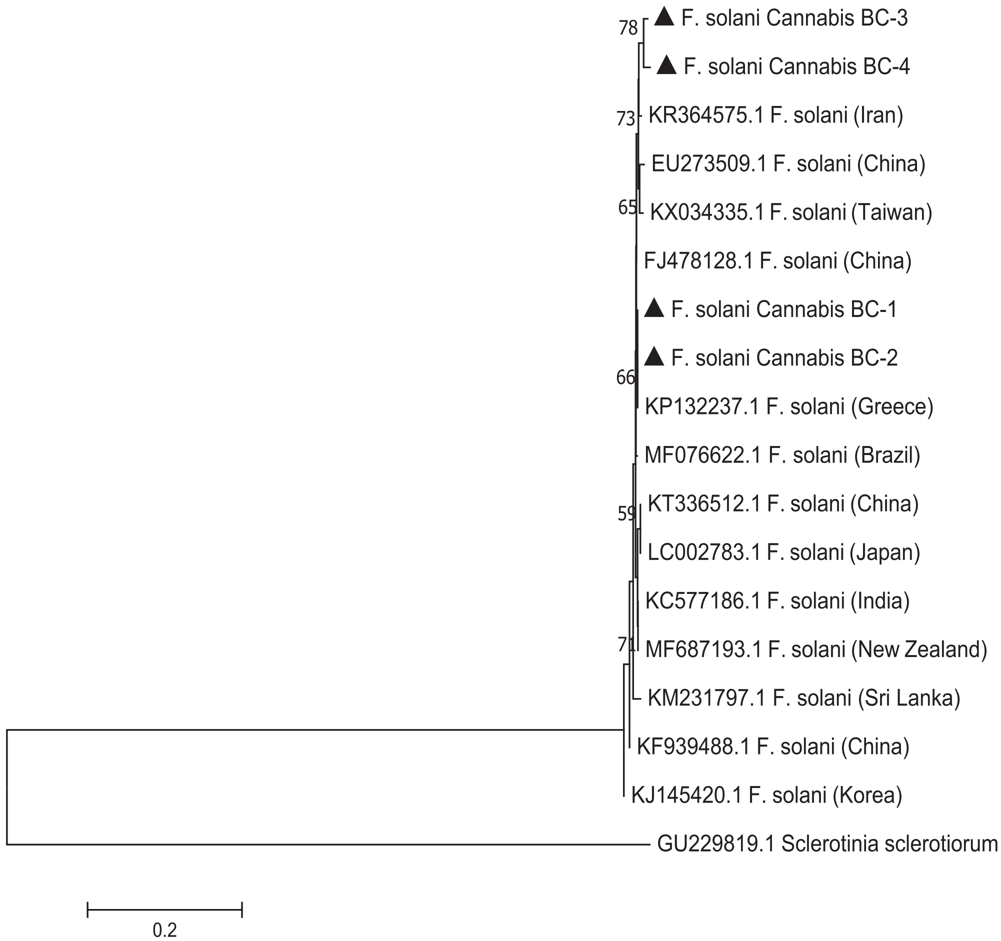


**Fig. 4** Phylogenetic tree of *Fusarium solani* isolates originating from diseased cannabis roots (BC-1, BC-2, from a previous study) and cannabis buds (BC-3, BC-4, present study) using ITS1-5.8S-ITS2 sequences compared with 13 isolates from a range of geographic regions worldwide (GenBank numbers are shown). A bootstrap consensus tree was inferred from 1000 replicates to represent the distance using the neighbour-joining (NJ) method. Branches corresponding to partitions reproduced in less than 50% bootstrap replicates were collapsed. *Sclerotinia sclerotiorum* was used as an outgroup. The scale bar indicates the expected number of nucleotide substitutions.

were prolific (Fig. 6c,d) and spores were frequently observed stuck to the glandular trichome surface (Fig. 6e,g). Germinating spores produced a germ tube (Fig. 6f) on the surface of bud tissues. The isolates BC-1 and BC-2 (GenBank accession nos. MH782036 and MH782037) showed 100% sequence identity in the ITS region to *Golovinomyces cichoracearum* accession no. EU233820.1 from orange coneflower (Garibaldi et al., 2008) and from KX897303.1 from chrysanthemum (Fig. 7). In addition, a 100% sequence match was also found with accession no. KM657962.1 from sunflower, previously identified as *G. ambrosiae* (Park et al., 2015), in addition to 100% sequence identity to *G. spadiceus* from dahlia, accession no. KX821733.1 (Fig. 7).

## Discussion

Among the diseases previously reported to affect hemp inflorescences grown under field conditions or indoors, the two most prevalent were grey mould and powdery mildew (McPartland, 1991, 1992). Botrytis bud rot (grey mould) caused by *B. cinerea* is a common problem encountered during cultivation of cannabis and causes a dark brown discolouration and internal rot of the buds. Grey mould was considered to be one of the most important diseases of cannabis, affecting flower buds and causing them to decay and also producing stem lesions (McPartland, 1992), particularly under conditions of high humidity. Cannabis strains with large inflorescences and tightly clustered flower buds that retained moisture were considered to be more susceptible to

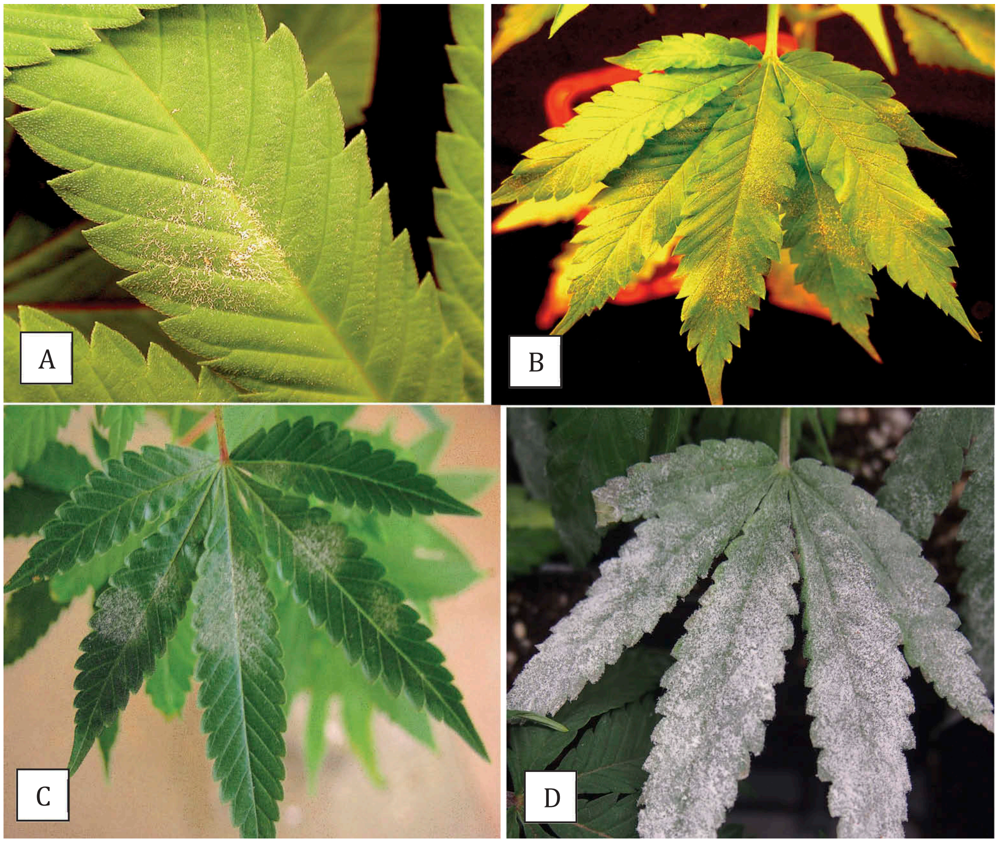


**Fig. 5** (Colour online) Symptoms and development over time of powdery mildew on cannabis leaves. **(a, b)** Young leaves; **(c, d)** Mature leaves. Characteristic white mycelium and sporulation are evident.

infection (McPartland, 1992). Approximately 2% of the flower buds collected in this study yielded *B. cinerea*.

Penicillium bud rot has not been previously described on cannabis and was shown in the present study to be caused by *P. olsonii* and a much lesser extent by *P. copticola*. The former species has also been reported to cause a post-harvest disease on greenhouse-grown tomato fruit (Chatterton et al., 2012), is a common inhabitant of decaying plant material and is present in air samples (Punja et al., 2016). The fungus was recovered from pre-harvest and post-harvest flower buds in this study. *Penicillium copticola* was previously reported to be present as an endophyte in twigs, leaves, and apical and lateral buds of cannabis plants (Gautam et al., 2013; Kusari et al., 2013). While it was suggested from dual culture antagonistic assays on agar media that *P. copticola* has biocontrol potential solely based on formation of antagonism zones when challenged against other fungi, our inoculation tests suggest that it can be a weak pathogen on cannabis buds. Although *P. copticola* has a less potentially negative effect on the quality of the buds than *P. olsonii*, it was recovered from surface-disinfested stems and petioles in this study and previously by Kusari et al. (2013), suggesting it is a prevalent endophyte. The occurrence of *P. copticola* within cannabis tissues can be problematic during plant tissue culture experiments (unpublished observations). Other prevalent *Penicillium* species previously detected on cannabis buds include *P. citrinum* and *P. paxillii* (Houbraken et al., 2010; McKernan et al., 2016a, 2016b). A more extensive sampling for *Penicillium* species potentially present on cannabis buds may reveal additional species to the two described in the present study.

Several *Fusarium* species were present on pre-harvest flower buds and *F. solani* was recovered from buds with brown rot symptoms that initially resembled grey mould. Both *F. solani* and *F. oxysporum* caused infection of the bracts and pistils following inoculation onto flower buds. Both species have also been previously reported to infect the roots of cannabis plants, causing root and crown rot (Punja & Rodriguez, 2018). While the *F. oxysporum* isolates from flower buds were found to be 100% identical in the ITS1-ITS2 region to those recovered

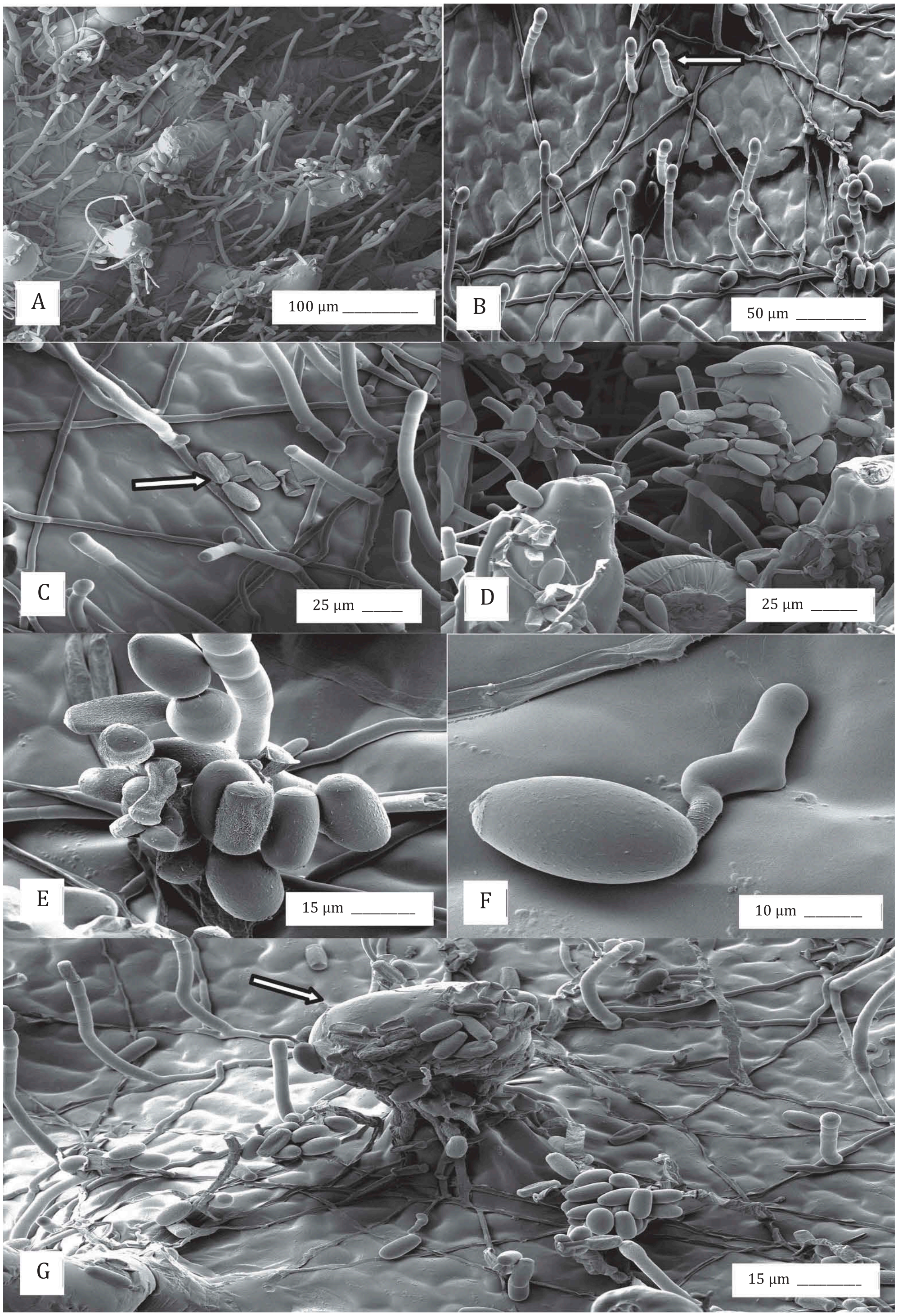


**Fig. 6** Scanning electron microscopy observations of the development of powdery mildew on cannabis buds. **(a, b)** Mycelium and conidiophores; **(c, d, e)** Conidial production; **(f)** Conidial germination; **(g)** Conidia and conidiophores on the bract surface and adherence of conidia to the surface of a trichome. Arrows show conidiophore **(b)**, mycelium **(c)** and glandular trichome head **(g)**.

previously from roots (Punja & Rodriguez, 2018) (data not shown), the *F. solani* isolates from buds (BC-3, BC-4) formed a separate subgroup from those recovered previously from roots (Punja & Rodriguez, 2018). *Fusarium solani* is frequently isolated from soil and plant debris and is a complex of over 45 phylogenetic

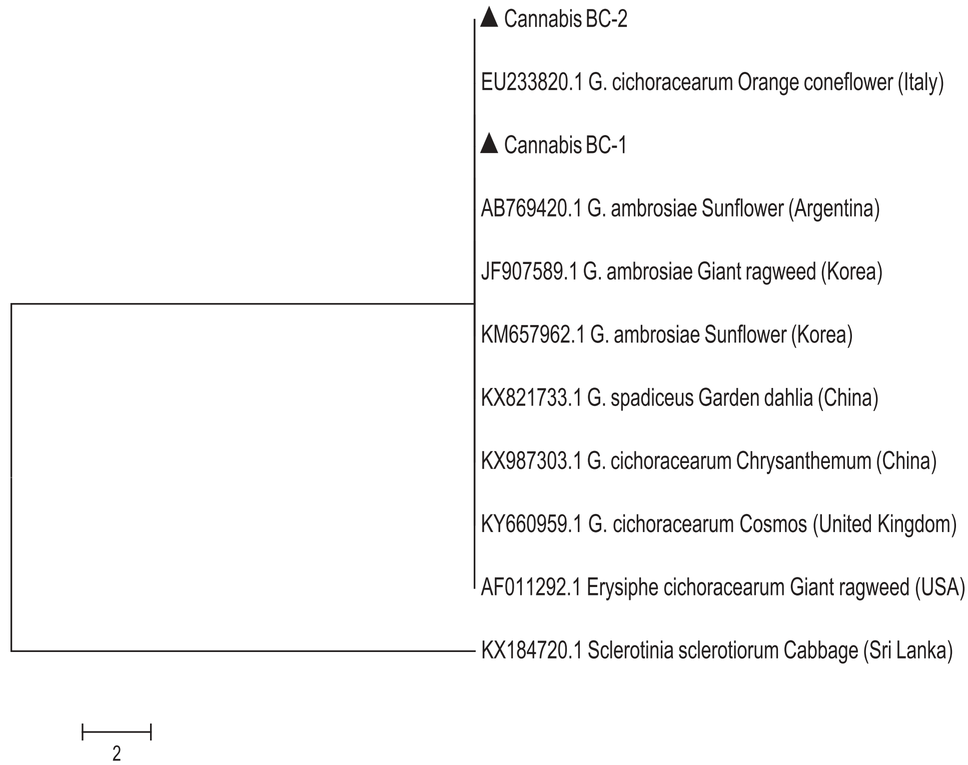


**Fig. 7** Phylogenetic tree of *Golovinomyces* species originating from cannabis (BC-1, BC-2) compared to those of other plant species using ITS1-5.8S-ITS2 sequences (GenBank numbers are shown). *Golovinomyces cichoracearum* could not be resolved using the ITS region from *G. ambrosiae* from sunflower and giant ragweed or *G. spadiceus* from dahlia. The scale bar indicates the expected number of nucleotide substitutions.

and/or biological species, termed the *Fusarium solani* species complex (FSSC). A phylogenetic study based on DNA sequences of three genes from 35 isolates indicated a high degree of phylogenetic diversity exists within this complex (O'Donnell, 2000). Isolates from flower buds in the present study originated from a different greenhouse than those from diseased roots sampled in a previous study (Punja & Rodriguez, 2018), suggesting there is genetic diversity among *F. solani* isolates originating from cannabis plants. In addition, both *F. oxysporum* and *F. solani* can be disseminated as airborne propagules (Gamliel et al., 1996; Katan et al., 1997) and sporulation was observed on cannabis buds in this study.

The sources of inoculum of the various bud rot pathogens described in this study likely include spores released from decomposing plant materials, on propagative plant materials, as well as airborne inoculum. In a previous study, Petri dishes placed in the growing environment of an indoor production facility revealed the development of both *Penicillium olsonii* and *Fusarium oxysporum* colonies, indicating that dissemination through the air was taking place (Punja & Rodriguez, 2018). Propagation of cannabis plants in greenhouses previously used to grow other crop plants which are susceptible to *Fusarium*, *Penicillium* and *Botrytis* pathogens, such as tomato and cucumber, may also result in introduction and spread of these pathogens. In the regulatory assessment of cannabis to fulfil quality assurance requirements, colony-forming units of total yeasts and moulds on dried buds are enumerated through dilution-plating on agar media, in addition to testing for potentially pathogenic bacteria (European Pharmacopoeia, 2015). In Canada, the maximum limit is $5 \times 10^4$ CFU of mould and yeast per g of dried product (Health Canada, 2013). The presence of *Penicillium* and *Fusarium* species on flower buds, as demonstrated in the present study, can potentially contribute to the total mould count, as well as potentially causing bud rot under conditions of high humidity or moisture during production or storage. Our method for recovery of the fungi involved placing individual flower buds in a plastic bag for 48 h prior to isolation. This was done to encourage fungal growth prior to isolation (enrichment step), but should not have affected the incidence of the fungi present, since each bud was placed aseptically in separate bags and data on incidence was recorded as present/absent.

Thompson et al. (2017), using DNA from cannabis samples collected from dispensaries in northern California, estimated fungal communities present in the microbiome by PCR of the internal transcribed spacer (ITS) gene sequence followed by metagenomic analysis. Among the genera present were the following, in decreasing intensity of detection: *Penicillium, Cladosporium, Golovinomyces, Aspergillus, Alternaria, Botryotinia, Chaetomium* and a low frequency of *Fusarium* (Thompson et al., 2017). Most of these fungi are common constituents of outdoor and indoor air samples (Meklin et al., 2007). Additional studies have reported the widespread presence of *Penicillium* and *Aspergillus* species as contaminants on cannabis buds (McPartland, 1994; Gautam et al., 2013; McKernan et al., 2016a, 2016b), as well as low detection of *F. oxysporum* (McKernan et al., 2016b). These fungi are commonly found in soil and on plant materials (Houbraken et al., 2010; Garba et al., 2017) and can also be found in the greenhouse environment (Gamliel et al., 1996; Katan et al., 1997; Punja et al., 2016; Punja & Rodriguez, 2018). While there is likely a higher diversity of fungi and potentially higher mould contamination rates on outdoor-produced cannabis compared to indoor-grown cannabis, this has not been conclusively shown in comparative studies. The overall incidence of fungi found on cannabis buds following isolation on PDA in this study was 27.5%. In the study by Thompson et al. (2017), they reported molecular detection (which could represent live and dead cells) of the following genera of fungi (frequency in 19 samples): *Penicillium* (18/19), *Cladosporium* (17/19), *Golovinomyces* (16/19), *Alternaria* (13/19) and *Fusarium* (6/19). The high frequency of *Penicillium* could be explained in part by its occurrence as a common endophyte (Gautam et al., 2013; Kusari et al., 2013) and by its prevalence in air samples. In contrast, McKernan et al. (2016b) reported the presence of mould and yeast in 6/17 dispensary-derived cannabis bud samples using qPCR, which also included several *Penicillium* species. One approach to reducing these fungal contaminants on cannabis buds is the prescribed use of post-harvest gamma-irradiation (Meklin et al., 2007; Hazekamp, 2016), which is currently being implemented by some licensed producers in Canada (https://aphria.ca/blog/3-facts-about-medical-cannabis-irradiation/).

Powdery mildew on hemp was previously reported to be caused by *Leveillula taurica* (Lev.) G. Arnaud and *Sphaerotheca macularis* (Wallroth:Fr.) Lind (McPartland, 1991, 1992) but the isolates from this study were identified to belong to the *Golovinomyces cichoracearum* species complex. In its broadest sense, *G. cichoracearum* includes powdery mildew fungi producing two-spored asci (Salmon, 1900), while in its most recent and strict sense, it applies to a widespread pathogen that infects host plants in the Asteraceae (lettuce, chicory, endive, aster) (Dhanvantari & Jarvis, 1985; Koike & Saenz, 1996; Lebeda & Mieslerova, 2011; Mork et al., 2011; Braun & Cook, 2012). Isolates infecting cannabis were identical to GenBank sequences deposited as *G. cichoracearum* but could not be distinguished using the ITS region from *G. ambrosiae* reported to infect sunflower and giant ragweed (Park et al., 2015) and *G. spadiceus* from dahlia (Fig. 7). Therefore, at the present time, the confirmation of the species of powdery mildew affecting cannabis, provisionally named *G. cichoracearum sensu lato* (Pépin et al., 2018), will require additional sequence comparisons of gene regions other than the ITS to confirm its identity. Conidia of *G. cichoracearum* are airborne and the pathogen is commonly found on both indoor and outdoor grown plant species (Dhanvantari & Jarvis, 1985; Koike & Saenz, 1996; Lebeda & Mieslerova, 2011; Mork et al., 2011; Braun & Cook, 2012). Incipient infections on propagative material on cannabis plants frequently remain undetected; therefore, the pathogen can be introduced and spread through infected plant material or potentially spread from other infected host species. The adherence of conidia to the glandular trichomes as observed through the SEM is likely due to the extrusion of sticky resin from trichomes as they mature (Small, 2017), which could also explain the high frequency of detection by PCR (84%) of *Golovinomyces* in bud samples by Thompson et al. (2017). This resin would also enhance adherence of other fungal spores, such as *Penicillium* species, as observed in the present study. The viability of these spores has not been determined.

Different strains of cannabis are likely to differ in susceptibility to the foliar and flower-infecting pathogens reported in this study. However, a comparative study of the response of these genotypes to the diseases affecting the crop has not been conducted and is an important aspect of disease management. Observations made over the duration of the present study suggest that disease incidence can be influenced by the cannabis strain cultivated. For example, 'Pennywise' and 'Girl Scout Cookies' are susceptible to powdery mildew while others e.g. 'Moby Dick', 'Pink Kush' and 'Hash Plant' may have increased susceptibility to grey mould (unpublished observations). Therefore, studies on the epidemiology of, and host response to, these pathogens should identify additional opportunities for disease management through cannabis strain selection. Very few biological control products are registered in Canada for management of foliar and flower-infecting pathogens of cannabis, and include Prestop WP (*Gliocladium catenulatum* strain J1446) and Rootshield WP (*Trichoderma harzianum* Rifai strain RRL-AG2) for control of grey mould. However, persistence of these biocontrol agents on the inflorescences could potentially contribute to the final mould count on the harvested product, making only early applications feasible. Additional research to identify more biorational strategies for disease management of foliar and flower-infecting pathogens on cannabis plants is urgently needed.

## Acknowledgements

Gina Rodriguez and Cameron Scott conducted some of the pathogenicity studies, Sarah Chen provided assistance with the molecular analyses and Darren Sutton conducted the scanning electron microscopy studies.

## Funding

This research was funded by Agrima Botanicals in conjunction with matching funds from the Natural Sciences and Engineering Research Council of Canada (NSERC), Collaborative Research and Development Grant.

## References

Braun U, Cook RTA. 2012. Taxonomic manual of the erysiphales (powdery mildews). CBS Biodiversity Series. 11:1–707.

Chatterton S, Wylie AC, Punja ZK. 2012. Fruit infection and postharvest decay of greenhouse tomatoes caused by *Penicillium* species in British Columbia. Can J Plant Pathol. 34:524–535.

Clarke RC. 1981. Marijuana botany: an advanced study: the propagation and breeding of distinctive cannabis. Oakland CA: Ronin Publishing.

Dhanvantari BN, Jarvis WR. 1985. Powdery mildew (*Erysiphe cichoracearum*) of greenhouse lettuce in Ontario. Plant Dis. 69:177.

European Pharmacopoeia (EP). 2015. Version7.0 – section 5.1.4. microbiological quality of non-sterile pharmaceutical preparations and substances for pharmaceutical use. Strasbourg: Council for Europe.

Felsenstein J. 1985. Phylogenies and the comparative method. Amer Naturalist. 125:1–15.

Gamliel A, Katan T, Yunis H, Katan J. 1996. Fusarium wilt and crown rot of sweet basil: involvement of soilborne and airborne inoculum. Phytopathology. 86:56–62.

Garba MH, Makun HA, Jigam AA, Muhammad HL, Patrick BN. 2017. Incidence and toxigenicity of fungi contaminating sorghum from Nigeria. World J Microbiol. 4:105–114.

Garibaldi A, Bertetti D, Frati S, Gullino ML. 2008. First report of powdery mildew caused by *Golovinomyces cichoracearum* on orange coneflower (*Rudbeckia fulgida*) in Italy. Plant Dis. 92:975.

Gautam AK, Kant M, Thakur Y. 2013. Isolation of endophytic fungi from *Cannabis sativa* and study their antifungal potential. Arch Phytopathol Plant Protect. 46:627–635.

Hazekamp A. 2016. Evaluating the effects of gamma-irradiation for decontamination of medicinal cannabis. Front Pharmacol. 7: 108–113.

Health Canada. 2013. Technical specifications for testing dried marihuana for medical purposes. [Accessed 2018 Feb 16]. https://www.canada.ca/en/health-canada/services/drugs-health- products/medical-use-marijuana/licensed-producers/guidance-document-technical- specifications-dried-marihuana-medical-purposes.html.

Houbraken JAMP, Frisvad JC, Samson RA. 2010. Taxonomy of *Penicillium citrinum* and related species. Fungal Div. 44:117–133.

Katan T, Shlevin E, Katan J. 1997. Sporulation of *Fusarium oxysporum* f. sp. *lycopersici* on stem surfaces of tomato plants and aerial dissemination of inoculum. Phytopathology. 87:712–719.

Koike ST, Saenz GS. 1996. Occurrence of powdery mildew, caused by *Erysiphe cichoracearum*, on endive and radicchio in California. Plant Dis. 80:1080.

Kusari P, Kusari S, Spiteller M, Kayser O. 2013. Endophytic fungi harbored in *Cannabis sativa* L. : diversity and potential as biocontrol agents against host plant-specific phytopathogens. Fungal Divers. 60:137–151.

Laate EA. 2012. Industrial hemp production in Canada. Alberta agriculture and rural development publication. [Accessed 2017 Sept 16]. http://www1.agric.gov.ab.ca/$department/deptdocs.nsf/all/econ9631.

Lebeda A, Mieslerova B. 2011. Taxonomy, distribution and biology of lettuce powdery mildew (*Golovinomyces cichoracearum* sensu stricto). Plant Pathol. 60:400–415.

McKernan K, Spangler J, Helbert Y, Lynch RC, Devitt-Lee A, Zhang L, Orphe W, Warner J, Foss T, Hudalla CJ, et al. 2016a. Metagenomic analysis medicinal *Cannabis* samples; pathogenic bacteria, toxigenic fungi, and beneficial microbes grow in culture-based yeast and mold tests. F1000Res. 5:2471. October 7. [Accessed 2018 Mar 15]. https://f1000research.com/articles/5-2471/v1.

McKernan K, Spangler J, Zhang L, Tadigotla V, Helbert Y, Foss T, Smith D. 2016b. Cannabis microbiome sequencing reveals several mycotoxic fungi native to dispensary grade Cannabis flowers. F1000Res. 4:1422. [Accessed 2018 Mar 15]. https://f1000research.com/articles/4-1422/v2.

McPartland JM. 1992. A review of *Cannabis* diseases. J Intern Hemp Assoc. 3:19–23. [Accessed 2016 Mar 28]. www.internationalhempassociation.org/jiha/iha03111.html.

McPartland JM. 1994. Microbiological contaminants of marijuana. J Intern Hemp Assoc. 1:41–44.

McPartland SM. 1991. Common names for diseases of *Cannabis sativa* L. Plant Dis. 75:226–227.

Meklin T, Reponen T, McKinstry C, Cho S-H, Grinshpun SA, Nevalainen A, Vepsalainen A, Haugland RA, LeMasters G, Vesper SJ. 2007. Comparison of mold concentrations quantified by MSQPCR in indoor and outdoor air sampled simultaneously. Sci Total Environ. 382:130–134.

Mork EK, Kristiansen K, Jorgensen HJL, Sundelin T. 2011. First report of *Golovinomyces cichoracearum* as the causal agent of powdery mildew on *Symphyotrichum novi-belgii* (*Aster novi-belgii*) in Denmark. Plant Dis. 95:228.

O'Donnell K. 2000. Molecular phylogeny of the *Nectria haematococca-Fusarium solani* species complex. Mycologia. 92:919–938.

Park MJ, Kim BS, Choi IY, Se C, Shin HD. 2015. First report of powdery mildew caused by *Golovinomyces ambrosiae* on sunflower in Korea. Plant Dis. 99:557.

Pépin N, Punja ZK, Joly DL. 2018. Occurrence of powdery mildew caused by *Golovinomyces cichoracearum sensu lato* on *Cannabis sativa* in Canada. Plant Dis. 102: PDIS-04–18-0586.

Punja ZK, Rodriguez G. 2017. Foliar and root pathogens of *Cannabis sativa* (marijuana) in British Columbia. Can J Plant Pathol. 39:373 (abstr.).

Punja ZK, Rodriguez G. 2018. *Fusarium* and *Pythium* species infecting roots of hydroponically grown marijuana (*Cannabis sativa* L.) plants. Can J Plant Pathol. 40:4. doi:10.1080/07060661.2018.1535466

Punja ZK, Rodriguez G, Chen S. 2017. Assessing genetic diversity in *Cannabis sativa* using molecular approaches. In: Chandra S, Lata L, ElSohly MA, editors. *Cannabis sativa* L. Botany and biotechnology. Berlin: Springer-Verlag; p. 395–418.

Punja ZK, Rodriguez G, Tirajoh A, Formby S. 2016. Role of fruit surface microflora, wounding, and storage conditions on post-harvest disease development on fresh market tomatoes. Can J Plant Pathol. 38:448–459.

Punja ZK, Scott C, Chen S. 2018. Root and crown rot pathogens causing wilt symptoms on field-grown marijuana (*Cannabis sativa* L.) plants. Can J Plant Pathol. 40:4. doi:10.1080/07060661.2018.1535470

Saitou N, Nei M. 1987. The neighbor-joining method: A new method for reconstructing phylogenetic trees. Mol Biol Evol. 4:406–425.

Salmon ES. 1900. A monograph of the erysiphaceae. Mem Torr Bot Club. 11:1–292.

Small E. 2017. Cannabis. A complete guide. Boca Raton FL: CRC Press.

Tamura K, Nei M, Kumar S. 2004. Prospects for inferring very large phylogenies by using the neighbor-joining method. Proc Nat Acad Sci USA. 101:11030–11035.

Tamura K, Peterson D, Peterson N, Stecher G, Nei M, Kumar S. 2011. MEGA5: molecular evolutionary genetics using maximum likelihood, evolutionary distance, and maximum parsimony methods. Mol Biol Evol. 28:2731–2739.

Thompson GR III, Tuscano JM, Dennis M, Singapuri A, Libertini S, Gaudino R, Torres A, Delisle JPM, Gillece JD, Schupp JM, et al. 2017. A microbiome assessment of medical marijuana. Clin Microbiol Infect. 23:269–270.